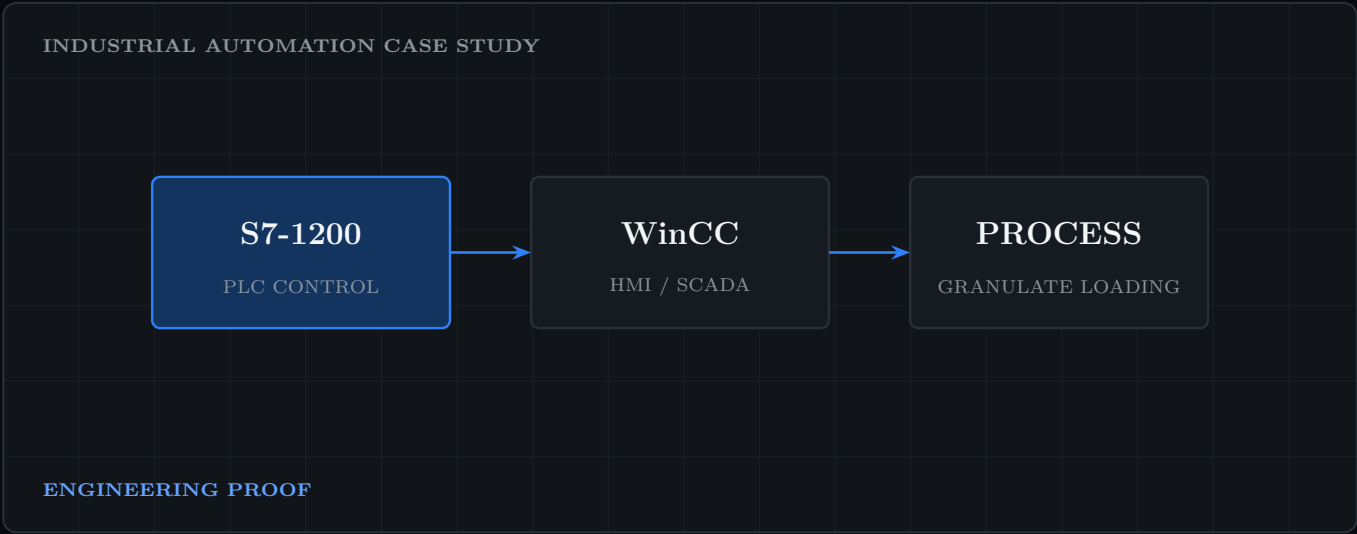


Automated Granulate Loading System

PLC-based automation and HMI supervision for an extrusion material-loading process.

Tunisie Câbles — One Tech Group



- Siemens S7-1200
- TIA Portal V18
- WinCC
- AUTOMGEN

From manual loading to automatic control

The initial process relied on manual monitoring of the material level and manual intervention when the reservoir became low.

INITIAL CONDITION Manual material loading The operator had to monitor the material level and refill the reservoir when the low-level indicator was activated. Consequences – Frequent production interruptions – Waiting time – Risk of human error	AUTOMATION OBJECTIVE Automate the loading cycle Automatically start material loading when the level becomes low and stop the process when the high level is reached. Integrated features – AUTO / MANU operation – Fault indication – HMI supervision – Safety conditions
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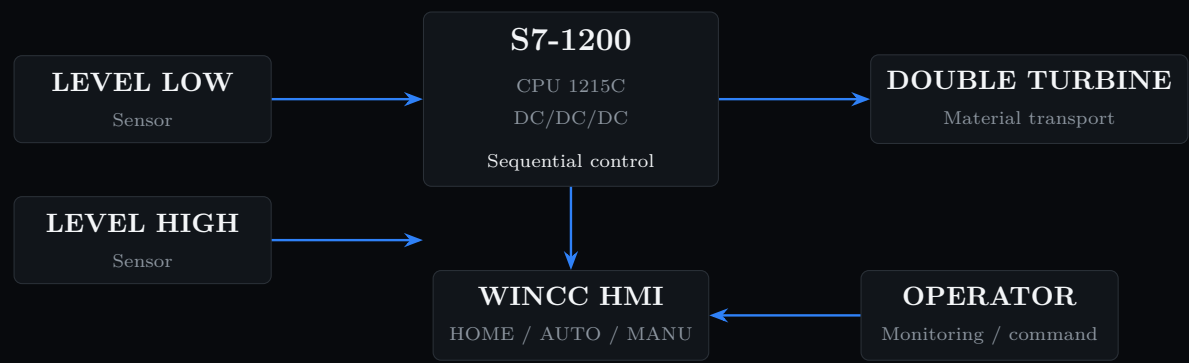
Functional concept



The PLC coordinates the sensors and actuator while WinCC provides the operator interface for visualization and command.

PLC-centered system architecture

The architecture separates process sensing, control logic, actuation and operator supervision.



TECHNICAL STACK	
PLC	Siemens S7-1200 CPU 1215C DC/DC/DC
PROGRAMMING	TIA Portal V18 — Ladder / functional blocks
SEQUENTIAL LOGIC	AUTOMGEN — Grafcet simulation and validation

Communication between the PLC and HMI enables process visualization and operator interaction through WinCC.

AUTO / MANU / SAFETY

The control strategy combines automatic material-level management, manual operation and safety interlocks.

Safety conditions satisfied ; **LOW LEVEL**
Material demand detected ; **LOADING**
Double turbine ON ; **HIGH LEVEL**
Filling condition reached ; **STOP LOADING**
Double turbine OFF ;

;;;

MODE 01

AUTO

The PLC manages the loading sequence according to the level sensors.

MODE 02

MANU

The operator can directly command the relevant actuators for testing and diagnosis.

MODE 03

SAFETY

Safety conditions override normal operation and inhibit unsafe states.

Safety interlocks

Emergency stop has priority over normal operation.

Overheating triggers an alarm condition.

AUTO and MANU modes are mutually exclusive.

Electrical protection includes fuses, thermal relay and grounding.

DESIGN PRINCIPLE

Normal control is always subordinate to the defined safety conditions.

The operator interface

WinCC provides three main HMI views for process monitoring, automatic operation and manual control.

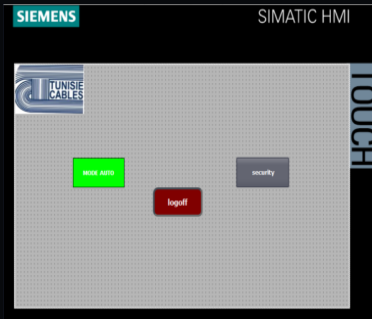


FIG. 01 HOME — system overview and status

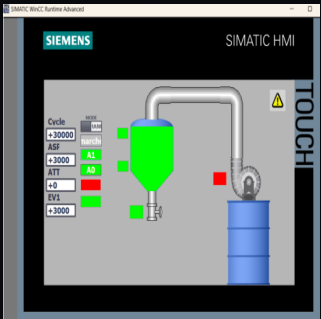


FIG. 02 AUTO — sequence and process states

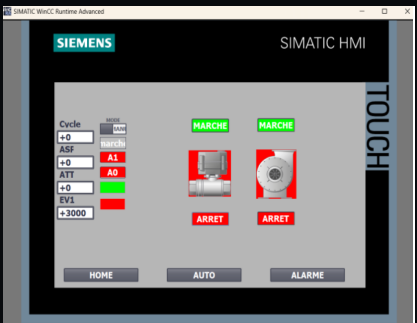


FIG. 03 MANU — direct control and diagnostics

HMI DESIGN LOGIC

HOME provides the global system view. **AUTO** exposes the automatic sequence and process states. **MANU** supports direct actuator control and diagnosis.

The interface also provides status information for sensors, outputs and fault conditions, supporting operator awareness and troubleshooting.

From process analysis to industrial automation

VALIDATION WORKFLOW

01 Process analysis → 02 Grafcet / logic → 03 TIA Portal

04 HMI → 05 Testing

AUTOMGEN + TIA PORTAL VALIDATION / REAL SYSTEM COMMISSIONING

01

PLC Programming

Understanding the role of a Siemens PLC in sequential industrial process control.

02

HMI / Supervision

Designing operator views for monitoring, automatic operation and manual diagnosis.

03

Sequential Automation

Translating a physical process into defined automatic and manual operating states.

04

Industrial Safety

Integrating emergency stop, overheating detection and inter-mode locking into the control strategy.

ENGINEERING SNAPSHOT

S7-1200 · TIA Portal V18 · WinCC · AUTOMGEN · Industrial Automation